

Python in Electrical Engineering

Fourier Transform

I. Example signal analysis

1. Open the files: signal_01.csv and signal_02.csv
2. Plot the signals in the time domain (clearly label axes and title)
3. Determine the frequency resolution (Δf) of the sampled signal
4. Compute the FFT and plot the magnitude spectrum
5. Identify and report the main frequency components present in the signals together with their approximate amplitudes

II. Example signal analysis

1. Open the file signal_02.csv
2. Plot the signal in the time domain
3. Calculate the frequency resolution (Δf)
4. Compute the FFT and plot the magnitude spectrum
5. Identify the main frequency components and their amplitudes
6. Discuss whether spectral leakage is present and make improvements to eliminate it

III. Demonstration of Aliasing Phenomenon

1. Generate the following signal using high time resolution (10k points or more).
 - 50 Hz component with amplitude 100 V
 - 950 Hz component with amplitude 30 V
 - Duration = 0.2 seconds
2. Sample this signal at two different sampling frequencies:
 - $f_{s1} = 5$ kHz (Proper sampling, above Nyquist)
 - $f_{s2} = 0.8$ kHz (Undersampling)
3. Plot the continuous signal and overlay the sampled points for both sampling rates (use two subplots)
4. Compute the FFT for both sampled versions and plot the magnitude spectra side-by-side.
5. Analysis Explain the observed differences between the two FFT plots. Specifically answer:
 - What happened to the 950 Hz component?
 - Why does this phenomenon occur?